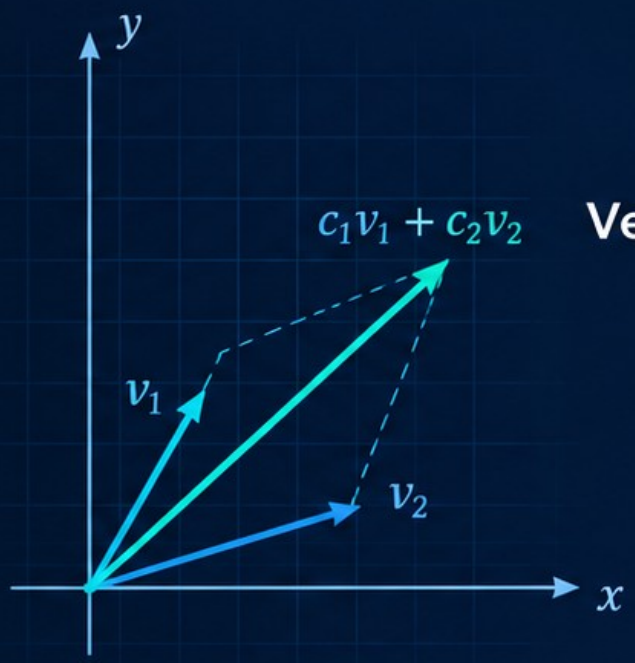
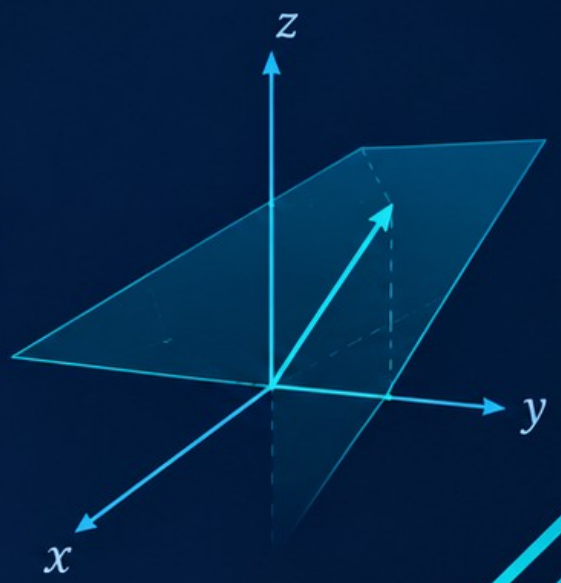
BETTER THINKING
BIGGER POSSIBILITIES

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

LINEAR ALGEBRA REVIEW





Vectors • Linear Combinations • Elimination • Matrices

HSIYC · Academic Support Department

Presented by Rahat

Today’s mission

Three ideas that connect into one method

2.1 VECTORS

A vector is a list of numbers.

A linear combination means:
scale vectors, then add them.

Question: Can we build a target vector?

2.2 ELIMINATION

Use one equation to cancel an unknown in another equation.

Make the problem smaller step by step.

Question: Which variable can disappear?

2.3 MATRICES

Write the same equations as rows of numbers.

Use legal row operations to create zeros.

Question: What operation creates the zero?

Big idea: Vectors → equations → matrix elimination → solution

2.1 | What is a vector?

A list of numbers with a clear order

Example

$$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Read it as: 2 across, 3 up.

Order matters: [2, 3] is not the same as [3, 2].

Think of an arrow

Start at the origin (0, 0).
Move 2 units right.
Move 3 units up.

The arrow ends at (2, 3).

A vector can describe direction and amount.

Quick check: How many entries are in a 3D vector?

2.1 | Linear combinations

Multiply each vector by a number, then add

Worked example

$v_1 = [1, 2]$ $v_2 = [3, 1]$

$2v_1 + 3v_2$
$= [2, 4] + [9, 3]$
$= [11, 7]$

Multiply EVERY entry. Then add matching positions.

Coefficients tell you how much

$2v_1$ means use two copies of v_1.

$-v_2$ reverses the direction of v_2.

A linear combination looks like:
$c_1v_1 + c_2v_2$

The c values are called coefficients.

PAIR TASK • Calculate 2[1, 3] + [2, 4].

2.1 | Can we build a target vector?

Linear combinations turn into equations

Question

Can these vectors build [7, 5]?

$c_1[1, 2] + c_2[3, 1] = [7, 5]$

Compare top entries:
$c_1 + 3c_2 = 7$

Compare bottom entries:
$2c_1 + c_2 = 5$

Solve the coefficients

From the equations:
$c_1 = 8/5,\ \ c_2 = 9/5$

Check by substituting them into the vector combination.

If the equations have no solution, the target cannot be built from those vectors.

Remember: “Can this target be built?” means “Can I solve for the coefficients?”

2.2 | The idea of elimination

Make one unknown disappear

Start with two equations

$x + y = 5$
$2x + y = 8$

Subtract the first equation from the second:
$(2x + y) - (x + y) = 8 - 5$

$x = 3$

Then substitute

Use $x + y = 5$.

$3 + y = 5$

$y = 2$

Answer: $x = 3$, $y = 2$

The y terms disappeared because $y - y = 0$.

Core trick: combine equations so one variable cancels to zero.

2.2 | Choose add or subtract

Look at the signs of the variable you want to remove

Same signs → subtract

$x + y = 7$
$2x + y = 10$

Subtract equation 1 from 2:
$x = 3$

Because $y - y = 0$.

Opposite signs → add

$x + y = 7$
$x - y = 3$

Add the equations:
$2x = 10$, so $x = 5$

Because $y + (-y) = 0$.

Ask yourself: What operation makes this chosen variable equal ZERO?

2.3 | Equations become a matrix

Rows keep the same information in a compact form

System

$$x + y + z = 6$$
$$2x + y + z = 8$$
$$x + 2y + 3z = 14$$

Each equation becomes one row.

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 8 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

Columns: x, y, z
Last column: right-hand side

The vertical bar separates coefficients from answers.

Never lose the final column: apply row operations to it too.

2.3 | The three legal row operations

They preserve the solution set

1 • Swap

$R_1 \leftrightarrow R_2$

Exchange two complete rows.

Useful when a better pivot is lower down.

2 • Scale

$R_2 \to 3R_2$

Multiply every entry in a row by the same nonzero number.

3 • Replace

$R_2 \to R_2 - 2R_1$

Take 2 × row 1 away from row 2; replace row 2 with the result.

Most common in elimination: row replacement.

Worked example | Create zeros

The goal is a staircase of pivots

Start

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 8 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -4 \\ 0 & 1 & 2 & 8 \end{array}\right]$$

Next

$R_3 \to R_3 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -4 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Now solve from the bottom:
$z = 4$, $y = 0$, $x = 2$

Solution: (x, y, z) = (2, 0, 4)

Pivots and back-substitution

Read the staircase from bottom to top

Pivot

A pivot is the leading nonzero entry in a row.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -4 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Pivots: $1, -1, 1$

Back-substitution

Bottom row: $z = 4$

Middle row: $-y - z = -4$
$-y - 4 = -4 \rightarrow y = 0$

Top row: $x + y + z = 6$
$x + 0 + 4 = 6 \rightarrow x = 2$

Tip: solve the easiest bottom equation first, then work upward.

What can elimination tell you?

Not every system has exactly one solution

One solution

A consistent system with a pivot for every variable often gives one unique solution.

Example:
x + y = 5
x − y = 1

No solution

A row like:
[0 0 | 5]

means 0 = 5, impossible.

The equations contradict each other.

Infinite solutions

A row like:
[0 0 | 0]

adds no new restriction.

A free variable may remain.

Interactive checkpoint

Pause, discuss with a partner, then reveal the answers

Try these without notes

1) 2[1, 3] + 3[2, 1] = ?

2) If the target entry is 4 and pivot is 1, what multiple of the pivot row cancels it?

3) What does [0 0 | 5] tell you?

4) Why must the RHS column change too?

Answers • reveal after trying

1) [8, 9]

2) Subtract 4 times the pivot row.

3) No solution: it says 0 = 5.

4) It is part of the equation, so row operations must preserve the whole equation.

If you got one wrong, explain WHY before moving on.

You are more prepared than you think

A final reminder before the midterm

“Great things are done by a series of small things brought together.”

— Vincent van Gogh

Midterm reminder

Read carefully • Show each row operation • Check signs • Verify your final answer

Wishing you all the very best for your midterm. Stay calm, think clearly, and trust your preparation.

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Questions after the review?

Keep practicing.
Ask questions when stuck.
Help a classmate when you can.

Best wishes for a confident and successful midterm!

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“Small steps. Clear thinking. Strong results.”

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